CHAPTER 24

QUANTUM PHYSICS

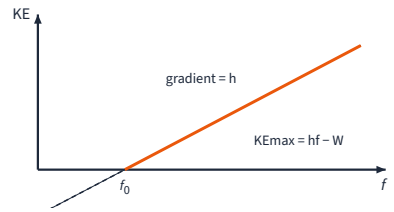
Key idea: light comes in photons of energy hf, and moving particles behave as waves of wavelength h/p.

1 PHOTON

$$E = hf = \frac{hc}{\lambda} \qquad p = \frac{h}{\lambda}$$

- A quantum (packet) of electromagnetic energy
- 1 eV $= 1.60 \times 10^{-19}$ J
- E (eV) ≈ 1240 ÷ λ (nm)

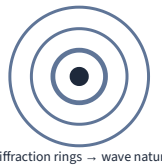
2 PHOTOELECTRIC EFFECT



$$hf = \phi + K_{\max} \qquad eV_s = K_{\max}$$

Graph of $K_{\max}$ against f: gradient h, f-intercept $f_0 = \phi/h$.

3 WAVE-PARTICLE DUALITY



electron diffraction rings → wave nature, λ = h/p

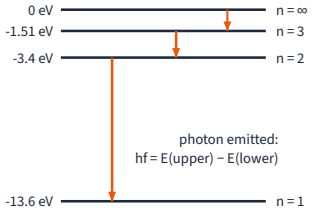
$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2meV}}$$

Electron diffraction rings show the wave nature of electrons.

4 WAVE THEORY FAILS

1. There is a **threshold frequency** f_0.
2. $K_{\max}$ depends on frequency, **not** intensity.
3. Emission is **instantaneous**.
4. Intensity changes only the number of photoelectrons.

5 ATOMIC STRUCTURE

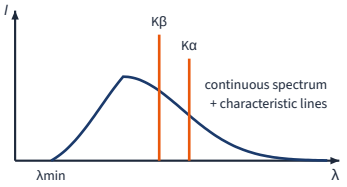


$$E_n = -\frac{13.6}{n^2}\ \text{eV}$$

$$hf = E_i - E_f$$

Lyman → $n = 1$ (UV); Balmer → $n = 2$ (visible); Paschen → $n = 3$ (IR). Bohr: $mvr = \dfrac{nh}{2\pi}$.

6 X-RAYS



$$\lambda_{\min} = \frac{hc}{eV} \qquad 2d\sin\theta = n\lambda$$

- Continuous spectrum: braking (bremsstrahlung)
- Sharp lines: characteristic of the target
- Bragg angle θ is measured from the **planes**

7 NANOSCIENCE

- Size 1–100 nm; huge surface-area-to-volume ratio ($3/r$ for a sphere)
- Quantum confinement: $E_n = \dfrac{n^2h^2}{8mL^2}$, smaller dot → bluer light
- STM: tunnelling current depends exponentially on the gap

8 COMMON SECTION A TRAPS

- Using eV in $hf = \phi + K$ while h is in J s.
- Saying brighter light gives faster photoelectrons.
- Forgetting energy levels are **negative**.
- Using the electron mass for a proton in $\lambda = h/p$.
- $\lambda_{\min}$ depends on the voltage, not the target.

EXAM TIPS

- ✓ Convert every energy to joules (or every one to eV) before substituting.
- ✓ Draw the transition arrow on the level diagram.
- ✓ For $\lambda_{\min}$, all the electron KE becomes one photon.
- ✓ Quote $h = 6.63 \times 10^{-34}$ J s.